

TrustShift: A Mechanism-Aware Audit of Machine-Learning Deployment Shift

A cross-domain protocol for discrimination, calibration, operating-point, and subgroup-reliability auditing

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Abstract

Deployment evaluation implicitly assumes that a larger distribution shift means greater risk; we find this does not hold uniformly across four real deployment shifts. We introduce **TrustShift**, a single protocol measuring discrimination, calibration, subgroup reliability, and operating-point performance, applied to clinical risk prediction (NHANES→BRFSS), mental-health text classification (Kaggle→Reddit/Twitter), mortgage-approval prediction (temporal and cross-state shift over 42M U.S. loan records), and network-intrusion detection (CIC-DDoS2019→CICIDS2017). A lending model under measured covariate shift (domain-classifier AUC up to 0.80) improves in ranking (AUC) while its macro-averaged operating-point performance stays essentially flat (within 0.02) even as its positive-class F1 alone falls by up to 0.078: a single metric does not tell the whole story, and shift magnitude alone did not predict which one would move. A mental-health model at comparable domain-classifier magnitudes loses up to 0.39 AUC. Cheap, largely label-free probes – prevalence change, a domain classifier, an importance-reweighting check – recover a mechanistic diagnosis for some shifts and report *inconclusive* for others when evidence is insufficient. A network-intrusion detector holds a near-ceiling in-domain AUC (0.998, capture-session-disjoint split) while collapsing to 0.68 on deployment traffic whose attack mechanisms are entirely absent from training. Post-hoc recalibration cuts calibration error by one to two orders of magnitude in every domain and generally leaves discrimination and subgroup reliability unrestored (security’s AUC moves only +0.035, short of recovering its 0.31 collapse). We contribute a protocol, a four-domain benchmark, and a release in which every reported number traces to a committed result file.

Keywords: distribution shift · deployment auditing · model calibration · subgroup reliability · shift diagnosis · trustworthy machine learning

1 Introduction

Consider three machine-learning systems at the moment of deployment. A clinical risk model moves from a survey cohort to a national screening population; its discrimination barely changes, and it looks fine—yet its probability estimates become substantially miscalibrated, so the risk numbers it reports are wrong. An intrusion detector moves from one year’s attack corpus to the next; its in-domain AUC sits near the ceiling (0.998, measured under a capture-session-disjoint train/test split), and it looks flawless—yet it collapses on deployment traffic whose attack mechanisms are entirely absent from its training data. A mortgage-approval model moves across years and states

under a distribution shift large enough to be detected almost perfectly by a classifier trained to spot it—its ranking of applicants (AUC) even improves, while its fixed-threshold approval decisions shift as the underlying approval rate moves.

Three systems, three intuitions overturned. We argue these are not isolated accidents but three faces of one deployment phenomenon. The one that looked fine failed; the one that looked perfect failed; the one that looked most shifted did not. Why? Answering that question is the subject of this paper, and the answer is not the magnitude of the shift alone.

Deployment evaluation is dominated by a single in-domain number—accuracy, or AUC—treated as evidence of readiness. When shift is considered at all, it is treated as a scalar severity: bigger shift, bigger risk, rather than as a mechanism with its own trustworthiness consequences. Clinical prediction has a mature external-validation literature; fairness-under-shift has been examined in medical imaging [Schrouff et al., 2022]; tabular shift has dedicated benchmarks [Gardner et al., 2023]; general shift benchmarks span vision and language [Koh et al., 2021]. Each answers “how much does accuracy drop under shift?” None answers the cross-domain question our three vignettes raise: *are deployment failures of trustworthiness domain-specific accidents, or do they follow rules that hold across domains—and can those rules be read before deployment?*

We answer it with one protocol applied to four deliberately dissimilar domains (the three above plus mental-health text classification). The protocol jointly measures discrimination, calibration, subgroup reliability, and fixed-threshold operating-point performance, attempts a staged, largely label-free mechanistic diagnosis of each shift, and quantifies what a remediation ladder restores. Holding the protocol fixed across clinical tabular data, social-media text, mortgage records, and network traffic separates what is general from what is idiosyncratic, and lets us ask directly: **how reliably can deployment-shift magnitude and context explain machine-learning failure fingerprints across trustworthiness dimensions, and how can mechanism-aware diagnostics guide deployment auditing?** Across these four domains, scalar shift-magnitude measures alone do not consistently explain which axis fails; deployment context, evaluated through a staged diagnostic protocol that explicitly reports *inconclusive* rather than forcing every shift into a category, carries information magnitude misses. Whether shift *mechanism* specifically outpredicts shift *magnitude* is a narrower question this paper tests directly and does not resolve in the affirmative.

Contributions.

- **A joint deployment-trustworthiness audit.** A single protocol measures discrimination, calibration, subgroup reliability, and fixed-threshold operating-point performance together rather than one axis at a time, so that axes shown to move independently are not missed by a single-metric evaluation (Section 2).
- **Evidence that scalar shift magnitude alone is insufficient.** Within the evaluated settings, scalar measures of marginal distribution change (a domain-classifier AUC, a prevalence shift) do not consistently explain which trustworthiness axis fails or by how much (Section 4.3); a lending model at large measured covariate shift stays robust in ranking while its fixed-threshold operating performance moves, and a mental-health model at a comparable domain-classifier magnitude collapses on every axis.
- **Deployment-context heterogeneity beyond magnitude.** Where prior audit history for a deployment population exists, that population’s identity explains substantially more of the observed failure variation than scalar shift magnitude does—evidence that deployment context, not only shift magnitude, is worth measuring before deployment.

- **A staged, largely label-free diagnostic protocol.** Cheap probes—prevalence change, a domain classifier, and an importance-reweighting check—recover a mechanistic diagnosis for some shifts and explicitly report *inconclusive* for others when the evidence does not support one, rather than forcing every shift into a category (Section 2).
- **A remediation ladder.** Across all four domains, post-hoc recalibration is the cheapest repair and reliably restores calibration (Section 4.5); it does not restore discrimination or subgroup reliability lost to shift, which require target supervision.
- **A mechanism-aware case study.** Network-intrusion detection shows severe, mechanism-specific degradation on attack families entirely absent from training, reported here as a supporting case study, not as general evidence that shift mechanism outpredicts shift magnitude.
- **A reproducible four-domain benchmark.** Harmonized shift-pairs, standardized predictions, and the protocol implementation are released so that every number in this paper, and any new domain placed on the same axes, regenerates from a committed result file.

2 The TrustShift Protocol

The protocol consumes, for each domain, a source model’s predictions on an in-distribution test split (*source*) and on one or more deployment splits (*target*), together with a subgroup label per example. It runs in three stages – a multi-axis trustworthiness audit, a staged shift diagnosis (label-free screening first, a labeled confirmatory probe only where target labels and adequate support overlap exist), and remediation – realized as six reported quantities.

Axis 1: Discrimination. AUC (macro one-vs-rest for multiclass domains), with DeLong 95% confidence intervals [DeLong et al., 1988]. We report $\Delta\text{AUC} = \text{AUC}_{\text{source}} - \text{AUC}_{\text{target}}$.

Axis 2: Operating-point performance. Macro-F1 at a fixed 0.5 threshold, reported alongside AUC because the two can move by different amounts or in different directions (Section 4). For binary domains we additionally report positive-class F1 where it diverges materially from macro-F1: macro-F1 averages both classes’ F1 equally, so it can mask a real, class-specific change that positive-class F1 alone reveals (Section 4.2).

Axis 3: Calibration. Expected Calibration Error (ECE, 10 equal-width bins) and Brier score per split. ECE quantifies the gap between predicted confidence and empirical accuracy [Guo et al., 2017]; a model can discriminate perfectly and still be badly calibrated, and vice versa.

Axis 4: Subgroup reliability. Per subgroup g , AUC_g ; the subgroup gap $G(\text{split}) = \max_g \text{AUC}_g - \min_g \text{AUC}_g$; and the *gap change* $\Delta G = G(\text{target}) - G(\text{source})$ —the headline reliability quantity—with a stratified bootstrap 95% CI ($N = 2000$). Subgroups are demographic where available (clinical: age, sex, race, BMI; lending: race), platform-derived proxy classes for text, and *operational* (attack family) for intrusion detection. We call this subgroup-conditional reliability and, for intrusion detection, never interpret it as demographic fairness (Section 7).

Table 1: The four deployment shifts. Subgroup semantics deliberately differ; the protocol definition is identical across domains, though data availability determines which Stage-2 probes return usable evidence in each (Table 2).

Domain	Source $\rightarrow$ Target	Modality	Subgroup axis	Shift mechanism
Clinical	NHANES $\rightarrow$ BRFSS	tabular	age/sex/race/BMI	population
NLP	Kaggle $\rightarrow$ Reddit, Twitter	text	proxy class	platform/style
Lending	HMDA 2020–21 $\rightarrow$ 2022–24; cross-state	tabular	race, income	temporal, geographic
Security	CIC-DDoS2019 $\rightarrow$ CICIDS2017	network flow	attack family	unseen attack-family deployment

Axis 5: Shift diagnosis. Each shift is decomposed with three probes, not all equally cheap. *Label shift*: change in class prevalence $\Delta\pi$ (labels only; label-free at the target). *Covariate shift*: a domain classifier trained to separate source from target examples on shared features, scored by out-of-fold cross-validated AUC (AUC_{dc} ; 0.5 = no shift, $\rightarrow$ 1 = severe) [Rabanser et al., 2019] (label-free). *Concept shift*: importance-weighting the source examples by out-of-fold domain-classifier odds (with a sampling-prior correction applied before clipping) and recomputing source AUC; if the reweighted source AUC stays near the source AUC while the target AUC is far lower, covariate shift alone cannot explain the drop [Lipton et al., 2018]. This probe requires target *labels* (to compute the target AUC it compares against) and is therefore not label-free; we additionally require the reweighting to have adequate source/target support overlap, measured by effective sample size (ESS), before treating its result as evidence – when ESS falls below 10% of the source sample, we report the probe as non-adjudicable rather than asserting a conclusion from it. When no probe produces adjudicable evidence above threshold for a target, or when the evidence for one mechanism is directly contradicted by another measurement (e.g. a large $\Delta\pi$ where the positive-class populations are measurably disjoint between source and target, contradicting label shift’s requirement that $P(X | Y)$ be stable), the diagnosis is reported as **inconclusive** rather than forced into a taxonomy branch.

Axis 6: Remediation ladder. On a 10% stratified target calibration split we apply temperature, Platt, and isotonic recalibration [Zadrozny and Elkan, 2002], then re-measure discrimination, calibration, and subgroup reliability on the remaining 90%. Operating-point performance is not currently re-measured post-remediation. This isolates what a cheap post-hoc fix restores from what requires target supervision.

Full metric definitions, seeds, and estimator choices are fixed in the released configuration; all reported numbers regenerate from committed prediction files. Algorithm 1 states the protocol as executed.

3 Domains and Data

We chose four domains that differ in modality, subgroup semantics, and shift mechanism, so that any shared pattern is unlikely to be an artifact of one setting (Table 1).

The clinical and NLP source models reuse published predictions (Pall and Yadav, 2026, a preprint under review, for NLP, evaluated on GoEmotions [Demszky et al., 2020] and a Twitter emotion corpus; a diabetes external-validation study on NHANES [National Center for Health Statistics, Centers for Disease Control and Prevention, 2020] and BRFSS [Centers for Disease Control and Prevention, 2022] for clinical), so no retraining occurs and the source numbers reproduce prior work. The lending and security analyses are new: for lending we train gradient-boosted

Algorithm 1 The TrustShift audit

Require: source predictions $(\hat{p}, y, g)_{\text{src}}$; target predictions $(\hat{p}, y, g)_{\text{tgt}}$; shared features $X_{\text{src}}, X_{\text{tgt}}$

Stage 1 — Evaluate the four trustworthiness axes

- 1: $\Delta\text{AUC} \leftarrow \text{AUC}_{\text{src}} - \text{AUC}_{\text{tgt}}$; ECE, Brier per split ▷ DeLong CIs; 10 bins
- 2: $\Delta\text{F1} \leftarrow \text{F1}_{\text{src}}^{\text{macro}} - \text{F1}_{\text{tgt}}^{\text{macro}}$ ▷ operating point, threshold 0.5
- 3: $\Delta G \leftarrow (\max_g - \min_g)\text{AUC}_g^{\text{tgt}} - (\max_g - \min_g)\text{AUC}_g^{\text{src}}$ ▷ subgroup gap; bootstrap CI

Stage 2 — Diagnose the shift type

- 4: $\Delta\pi \leftarrow$ prevalence change; $\text{AUC}_{\text{dc}} \leftarrow$ out-of-fold CV-AUC of a source-vs-target classifier on X
- 5: $w(x) \leftarrow$ prior-corrected, out-of-fold domain-classifier odds; $\text{ESS} \leftarrow (\sum w)^2 / \sum w^2$
- 6: evidence $\leftarrow \emptyset$
- 7: **if** $\text{ESS}/n_{\text{src}} \geq 10\%$ **and** $\text{AUC}_{\text{src}}^{\text{rw}} \approx \text{AUC}_{\text{src}} \gg \text{AUC}_{\text{tgt}}$ **then** evidence $\leftarrow$ evidence $\cup \{\text{concept}\}$
- 8: **end if**
- 9: **if** $|\Delta\pi| > 0.15$ **then** evidence $\leftarrow$ evidence $\cup \{\text{label}\}$
- 10: **end if**
- 11: **if** $\text{AUC}_{\text{dc}} > 0.65$ **then** evidence $\leftarrow$ evidence $\cup \{\text{covariate}\}$
- 12: **end if**
- 13: **if** label $\in$ evidence **and** positive-class populations measurably disjoint **then** diagnosis $\leftarrow$ inconclusive
- 14: **else if** evidence $= \emptyset$ **then** diagnosis $\leftarrow$ inconclusive
- 15: **else if** $|\text{evidence}| = 1$ **then** diagnosis $\leftarrow$ the one mechanism in evidence
- 16: **else** diagnosis $\leftarrow$ mixed
- 17: **end if**

Stage 3 — Remediate and re-measure

- 18: fit recalibration on 10% target labels; re-measure discrimination, calibration, and subgroup reliability on the rest ▷ operating point not re-measured

Ensure: (ΔAUC , ΔF1 , ECE, ΔG , diagnosis, restored axes)

approval classifiers with no race or geographic feature on HMDA loan records [Consumer Financial Protection Bureau, 2024] and evaluate temporal (train 2020–21, test each of 2022/2023/2024) and geographic (train on 60% of states, test on held-out states) shift over 42.3M records; for security we train intrusion detectors on CIC-DDoS2019 [Sharafaldin et al., 2019] and evaluate on CICIDS2017 [Sharafaldin et al., 2018]. Data are public. Clinical target-side subgroup analysis is limited to age for a reason we state plainly in Section 7.

4 Results

We organize results as one argument: in-domain metrics do not predict deployment behaviour (4.1); each domain exhibits a distinct failure fingerprint, and the available shift evidence supports different levels of mechanistic interpretation across targets, with three of five inconclusive under the current probes (4.2); scalar shift magnitude does not explain that variation, while deployment context does, out of sample, for populations with prior audit history (4.3); a pooled aggregate can even reverse the sign of what most individual deployment cells show (4.4); and recalibration reliably repairs calibration but not discrimination or subgroup reliability (4.5).

Table 2: Stage-2 probe availability per domain (PLAN_ijdsa.md T1.6). A checkmark means the probe returned real, adjudicable evidence for at least one target in that domain; a dash means it was not run or never produced evidence above threshold. No domain has all three probes fail silently into a default diagnosis – **inconclusive** is reported explicitly instead (Table 2).

Domain	Label-shift ($\Delta\pi$)	Covariate (AUC_{dc})	Concept (reweighting)
Clinical	✓	–	–
NLP	✓	✓	✓
Lending	✓	✓ (temporal only)	–
Security	✓	–	–

4.1 In-domain performance does not transfer, and does not transfer uniformly

Every source model looked deployable: source AUC ranged from 0.77 (clinical) to 0.998 (security, measured under a capture-session-disjoint train/test split) and source ECE was at most 0.26. Deployment outcomes did not follow (Figure 2, Table 4). Mental-health classifiers lost 0.28–0.39 AUC on new platforms; the intrusion detector fell from AUC 0.998 to 0.684; the clinical model’s ranking discrimination moved only slightly ($\Delta\text{AUC} = 0.031$), though its operating-point performance (macro-F1) fell by 0.102; and lending models improved in ranking (AUC) on later years while their macro-averaged operating-point performance stayed essentially flat (change within 0.02) even as positive-class F1 alone fell by 0.056–0.078 as approval prevalence shifted. A single in-domain AUC dashboard would have rated all four “ready” and would have missed the operating-point divergence in clinical and, on the positive-class-F1 reading, in lending.

4.2 Shift evidence and failure fingerprints

The four domains exhibit distinct failure fingerprints (Table 3), while the available shift evidence supports different levels of mechanistic interpretation across the five evaluated targets: NLP carries **mixed** evidence, security and clinical are **inconclusive** under the current probes, lending temporal diagnoses as **covariate**, and lending geographic is **inconclusive**. For each target below we report the observed fingerprint first, the available diagnostic evidence second, and the resulting confidence level third.

Every axis fails together $\rightarrow$ evidence is mixed (label + covariate, and concept where adjudicable). The NLP shift’s diagnosis is **mixed**, not a single mechanism: the domain classifier separates platforms almost perfectly ($\text{AUC}_{\text{dc}} = 0.93\text{--}0.99$) and class prevalence shifts sharply ($|\Delta\pi|$ up to 0.52), so both covariate and label evidence fire for both targets. The concept-shift reweighting probe additionally fires for Reddit, where source/target support overlap is adequate (effective sample size 16% of n_{source}): reweighting barely changes source AUC ($0.990 \rightarrow 0.989$) while target AUC is ≈ 0.60 , so covariate shift alone cannot explain that drop. For Twitter, support overlap is too thin (effective sample size 4%) for the reweighting probe to adjudicate, and we report it as such rather than asserting a mechanism from it. Whichever probes fire, the outcome is the same: discrimination collapses (ΔAUC 0.28–0.39 across the four architectures tested), calibration breaks (target ECE up to 0.27), and the subgroup gap widens (ΔG up to +0.15). All three axes fail together, for the source-selected primary model (RoBERTa) and every other model tested.

Discrimination collapses on attack mechanisms absent from training $\rightarrow$ inconclusive by the current probes. The intrusion detector holds a near-ceiling in-domain AUC of 0.998, measured on a train/test split that is disjoint at the capture-file level (the dataset’s own native “-training”/“-testing” file partition) rather than a random row split. The native file/session split showed no

evidence of excess cross-split duplication relative to within-split duplication under our duplicate check: the across-split near-duplicate rate (2.0%) was no higher than the rate already present *within* the training files themselves (5.9%) or the testing files (2.2%); we do not claim this proves the split leakage-free, only that this specific check found no evidence of excess leakage. Under this evaluation, AUC still falls to 0.684 externally, and calibration error rises from 0.069 to 0.646. This is *not* label shift in the classical sense: raw (uncapped) class prevalence does shift substantially ($\Delta\pi = 0.633$, source 77% attack, target 14% attack – a larger shift than the training- pipeline’s engineered class balance would suggest), but label shift’s requirement that the positive-class conditional distribution $P(X | Y=1)$ stay stable is directly contradicted: all five attack families present at deployment are entirely absent from the source training data (see the label-space audit released with this benchmark), so the diagnosis is reported as **inconclusive** rather than forced into a label-shift label. Per-family recall shows a family that shares a label string with a training-side family (**ddos**) detected at only 0.37, alongside three families with no training-side counterpart at all detected near zero (**goldeneye** 0.00, **hulk** 0.0003, **slowloris** 0.21, **slowhttp** 0.07). We describe this observed pattern as an *unseen attack-family deployment shift*: the binary task (benign vs. attack) is unchanged, but the attack mechanisms present at deployment were never represented in training. Whether this is best formalized as a distinct taxonomy category alongside covariate/label/concept shift, or as a mixed/compound shift, is left to the controlled mechanism-isolation experiment in future work; we do not assign it a firm category here.

Ranking is robust, the operating point is nearly flat $\rightarrow$ evidence is covariate. The lending temporal shift diagnoses as covariate (AUC_{dc} 0.69–0.80) with no label or concept component. Ranking discrimination *improves* (ΔAUC -0.012 to -0.028) and calibration is stable; macro-averaged operating-point performance stays essentially flat (change within 0.02) even as positive-class F1 alone falls by 0.056–0.078 as the approval prevalence moves ($\pi = 0.850 \rightarrow 0.759$) and the denied class becomes less rare, and the race gap shrinks. The geographic experiment shows a different pattern: ranking discrimination also improves ($\Delta\text{AUC} = +0.059$) and the race gap narrows (-0.029), but macro-averaged operating-point performance degrades more than in the temporal experiment (macro-F1 falls by 0.034), even though its own shift is diagnosed **inconclusive**: no covariate probe has been built for the geographic target (see Table 2), so the shared ranking-improvement pattern is a numerical similarity to the temporal result, not a confirmed shared mechanism, pending the within-domain analysis in Section 7.

Calibration fails despite near-flat discrimination $\rightarrow$ evidence is inconclusive. The clinical shift’s diagnosis is **inconclusive**: the only available covariate probe for this domain is a score-distribution proxy (target-side features are unrecoverable, Section 7) that does not produce an AUC_{dc} , so no probe reaches the evidence threshold. The observed failure is nonetheless clear: the source-selected primary model (FedAvg) shows a small but statistically detectable ranking-discrimination shift ($\Delta\text{AUC} = +0.031$, ≈ 7.2 source-test standard errors) that is nonetheless 3–4 $\times$ smaller than its calibration and operating-point shifts: operating-point performance falls sharply (macro-F1 -0.102) and calibration deteriorates sharply (ECE 0.168 $\rightarrow$ 0.276; 0.41 for the non-primary boosted model). The age subgroup gap narrows for the primary model ($\Delta G = -0.066$, 95% CI $[-0.113, -0.016]$) but *widens* for the non-primary boosted model ($\Delta G = +0.029$, 95% CI includes zero) – subgroup reliability here is a property of the (model, shift) pair, not of the dataset alone. A model whose AUC certifies it as unchanged is producing risk estimates that are substantially miscalibrated, at a fixed operating point that has also moved.

Table 3: **The failure fingerprint.** One row per domain (source-only primary model, Section 2); lending has two rows because temporal and geographic shift are different experiments, not competing models. $\uparrow/\downarrow$ mark the direction and $\approx$ near-invariance. Discrimination is AUC (ranking); Operating point is macro-F1 at threshold 0.5 – the two are reported separately because they can disagree (Lending, Clinical). Diagnoses of **inconclusive** mean no probe reached the evidence threshold, or (Security) the one probe that fired is contradicted by other measured evidence – not that the shift is absent. *Macro-averaged operating-point performance is flat for lending temporal, but its positive-class F1 alone falls 0.06–0.08 (Section 4.2).

Domain	Diagnosis	Discrimination	Operating point	Calibration	Subgroup reliability
NLP	mixed	collapses ($\downarrow$ 0.28–0.39)	collapses ($\downarrow$)	breaks	gap widens ($\uparrow$)
Security	inconclusive	collapses (0.998 $\rightarrow$ 0.684)	collapses ($\downarrow$)	breaks ($\uparrow$ 9 $\times$)	unseen-family blind
Clinical	inconclusive	degrades less ($\downarrow$ 0.03)	falls ($\downarrow$ 0.102)	breaks silently ($\uparrow$)	model-dependent
Lending (temporal)	covariate	holds/improves ($\approx$)	flat* ($\approx$)	stable ($\approx$)	gap narrows ($\downarrow$)
Lending (geographic)	inconclusive	improves ($\uparrow$)	falls ($\downarrow$ 0.03)	stable ($\approx$)	gap narrows ($\downarrow$)

Table 4: Master results, source-only primary models (Section 2). Source/target AUC (ranking) and ECE, discrimination change, operating-point change (Δ macro-F1, positive = degrades), and subgroup-gap change ΔG (positive = gap widens under shift). Every number regenerates from committed prediction files; the complete table for every model tested (not only the primary) is released as `T2_master.csv`.

Domain	Target	AUC _{src}	AUC _{tgt}	Δ AUC	Δ F1	ECE _{src}	ECE _{tgt}	ΔG
Clinical (FedAvg)	BRFSS	0.788	0.757	+0.031	+0.102	0.168	0.276	−0.066
NLP (RoBERTa)	Reddit	0.987	0.637	+ 0.349	+0.580	0.029	0.112	+0.068
NLP (RoBERTa)	Twitter	0.987	0.603	+ 0.384	+0.599	0.029	0.258	+0.108
Lending (temporal)	2023	0.793	0.821	−0.028	−0.014	0.258	0.240	−0.031
Lending (geographic)	held-out states	0.845	0.786	+0.059	+0.034	0.206	0.190	−0.029
Security (XGBoost)	CICIDS2017	0.998	0.684	+ 0.314	+0.704	0.069	0.646	—

4.3 The regularity is categorical, not a magnitude law

Treating each (domain, model, target) as a shift-point ($n = 16$), we asked whether the subgroup-reliability change is predictable from cheaper quantities. *Because the observations cluster strongly by domain (effectively four clusters), we interpret every regression in this section as an exploratory cross-domain summary, not a fitted predictor.* With that caveat stated first: across shift-points, gap change is associated with calibration change (slope 0.42, 95% CI [0.17, 0.60], $R^2 = 0.41$) and with aggregate accuracy loss (slope 0.34, CI [0.27, 0.44], $R^2 = 0.82$). The tempting third association—subgroup loss with covariate-shift magnitude—has a high cross-domain fit (slope 0.54, $R^2 = 0.77$) but is a statistical trap, which we report against ourselves.

Unexpected finding: shift magnitude is not deployment damage

The cross-domain regression of subgroup-gap change on covariate-shift magnitude (AUC_{dc}) has slope $+0.54$ with $R^2 = 0.77$ —a fit that, taken at face value, would license “measure the shift; the bigger it is, the more subgroup reliability you lose.” That reading is wrong. Within the lending domain, where covariate-shift magnitude rises from 0.69 to 0.80 across deployment years, the same slope is *negative* (-0.10): more measured shift, less damage. The positive cross-domain slope is a two-cluster, aggregation-induced artifact separating high-shift text from moderate-shift tabular data—an aggregate line drawn through two groups that individually tell the opposite story. Lending exhibits larger measured covariate shift than many settings described as “severe,” while remaining fully robust on every axis measured this way. This mattered to our own analysis: an earlier version of this work reported the 0.54 slope as a finding before the within-domain check exposed it, and we retain it here precisely because the mistake is instructive. **Shift magnitude and deployment risk are not interchangeable**; the *categorical* diagnosis (Figure 1) provides a more structured description of the observed deployment evidence, but the present data do not establish that mechanism categorization predicts failure more accurately than scalar shift magnitude – that comparative question remains unresolved.

We therefore report the cross-domain regressions (Figure 3) as exploratory synthesis over four domain-clusters, not as a fitted predictor, and rest the contribution on the magnitude-insufficiency finding, not on a claim that mechanism categorization outpredicts it.

Deployment context explains far more of this variation than magnitude does, out of sample. If scalar shift magnitude does not explain the within-lending variation above, what does? Using the same 447-cell lending-temporal panel, we tested whether a state’s own identity – under 10-fold cross-validation, never trained and evaluated on the same fold – predicts its held-out cells better than the scalar shift score does. It does, by a wide margin, on every axis: state identity reaches $R^2_{out} = 0.62$ – 0.84 (discrimination 0.77, operating point 0.81, calibration 0.80, subgroup gap 0.62), while AUC_{dc} under the identical cross-validation scheme reaches at most $R^2_{out} = 0.06$ and is negative on three of six axes. This is context evidence, not mechanism evidence – shift mechanism is held constant (covariate) throughout this panel – and it comes with a sharp operational caveat that must be stated alongside it, not separately: the state *identifier* is available before any deployment, trivially, but its *predictive value* is not. Holding out an entire state never seen in training instead, state identity’s R^2_{out} collapses to approximately zero (-0.03 to -0.02), exactly as a fixed-effect encoding must for a level it has never observed. Deployment-context heterogeneity is therefore a real, generalizing, and substantial signal a scalar shift score misses – but only for a population with prior audit history, not as a universal pre-deployment diagnostic (`state_identity_generalization.json`, `context_generalization_final.json`).

4.4 Aggregation-induced reversal: a worked methodological lesson

The lending-temporal panel also illustrates a pitfall this paper’s own thesis warns against: an aggregate number can disagree with what is happening in the cells that compose it, in a way that would mislead an auditor who checks only the aggregate. Pooling all lending-temporal predictions for a target year and computing one macro-F1 delta – the whole-year aggregate reported in Section 4.2 – shows operating-point performance essentially flat to slightly improving (-0.003 to -0.019 across years). Computed the other way, per (state, year, seed) cell and averaged with every

Table 5: Remediation ladder (isotonic recalibration on 10% target labels), source-only primary models. ECE falls sharply in every domain; AUC is essentially unchanged except security.

Domain	Target	ECE (no fix)	ECE (isotonic)	AUC (no fix)	AUC (isotonic)
Clinical (FedAvg)	BRFSS	0.276	0.002	0.757	0.757
Lending (temporal)	2023	0.240	0.002	0.821	0.821
NLP (RoBERTa)	Twitter	0.260	0.062	0.594	0.592
Security (XGBoost)	CICIDS2017	0.646	0.004	0.684	0.718 [†]

[†] AUC moves by +0.035 under isotonic recalibration; whether this reflects a tie artifact (as it did for the previously-reported lightgbm result) has not been re-verified for this model and is noted as an open question, not restored discrimination.

cell counted equally, the picture flips: 66–80% of the 49–50 cells in each target year individually *degrade* (unweighted mean effect +0.024 to +0.046, median effect +0.025 to +0.056), yet the pooled aggregate improves in all three years. The reconciliation is arithmetic, not mysterious: weighting each cell by its row count reproduces the pooled aggregate to within 0.005–0.008, because a handful of large states (FIPS 12/Florida, 6/California, 48/Texas) dominate the row-weighted total while an unweighted per-state mean gives a state with a thousand loans the same vote as a state with a million. We call this an **aggregation-induced reversal** rather than Simpson’s paradox: the sign reversal itself is confirmed, but the classical two-group and lurking-variable conditions for that specific term were not formally verified here (`lending_aggregation_effect_final.json`). The practical lesson mirrors this paper’s central claim at a smaller scale: a single summary number – here, one pooled aggregate delta – does not tell the whole story, and an auditor who checks only that figure for a lending model would conclude operating-point performance is stable while a majority of the populations it serves are individually seeing it degrade.

4.5 Recalibration repairs calibration only

Isotonic recalibration on a 10% labeled target split reduced ECE by one to two orders of magnitude in every domain: clinical 0.276 → 0.002, lending 0.240 → 0.002 (temporal), security 0.646 → 0.004, NLP 0.111/0.260 → 0.020/0.062 (Table 5; source-only primary models throughout, Section 2). In clinical, lending, and NLP, AUC and the subgroup gap moved by less than 0.012—calibration is cheaply repairable, discrimination and reliability are not. Security is the exception: isotonic recalibration moves AUC by +0.035 (0.684 → 0.718). We do not yet have a rows-level tie count for the current (leakage-controlled, XGBoost) pipeline as we did for the earlier lightgbm result, so we report this as an open question rather than repeat the earlier “tie artifact” explanation unverified for this model; either way, +0.035 does not recover the 0.31 collapse. Restoring discrimination requires target supervision—e.g., in the NLP domain, target fine-tuning recovers far more than any post-hoc fix [Pall and Yadav, 2026].

5 Discussion

What failed, and why. The four domains produced four different failure patterns. NLP’s shift carries mixed evidence (both label and covariate probes fire, and the concept probe additionally fires for one target), and every axis failed together: discrimination, calibration, and subgroup reliability all collapsed. Security’s shift is diagnosed **inconclusive** by the current probes, yet discrimination and calibration collapsed together on the attack families absent from training, and

even the one family shared with training degraded substantially (Section 4.2). Lending’s temporal shift, diagnosed **covariate**, degraded on none of the four measured axes; its geographic shift, also evaluated here but diagnosed **inconclusive**, degraded operating-point performance while ranking still improved. Clinical’s shift, likewise inconclusive, affected calibration and the operating point far more than discrimination. Which axes failed together, and how severely, differed by target; whether that pattern is explained by shift *mechanism* specifically, rather than by deployment context or another factor the evaluated probes do not isolate, is the open question Section 4.3’s own direct test does not resolve in the affirmative.

The most practically important observation. Maintaining high discrimination after deployment did not imply trustworthy behaviour. The clinical model’s ranking discrimination moved only slightly while its probability estimates became substantially miscalibrated, and the lending model stayed robust or improved on every measured axis despite measurable covariate shift. These contrasting outcomes argue that deployment decisions should not rest on aggregate discrimination alone, but should incorporate the multi-axis audit this protocol runs—which is cheap and, for the covariate and label probes specifically, requires no target labels.

What practitioners should do differently. The results support a concrete pre-deployment routine, not a formally validated predictive decision rule: (1) screen with cheap, largely label-free probes – prevalence change, a domain classifier, and, where target labels exist, the reweighting check; (2) characterize what evidence is actually available, since not every target supports every probe (Table 2); (3) audit all four trustworthiness axes rather than aggregate discrimination alone; (4) report *inconclusive* honestly when the evidence does not support a firm diagnosis, rather than forcing one; (5) budget a small labeled target sample for recalibration, which was cheap and consistently improved calibration; and (6) escalate to target supervision when discrimination or subgroup reliability remain degraded after remediation, since recalibration alone did not reliably restore them. Monitoring aggregate AUC alone, the most common practice, was the least informative signal in all four domains.

Outlook. If this pattern continues to hold across additional domains, deployment evaluation may need to shift its emphasis from measuring *how large* a distribution shift is toward auditing multiple trustworthiness axes directly, since scalar magnitude alone did not consistently explain which axis failed in the settings studied here. Some shift evidence can be screened before deployment using unlabeled target data, while other mechanisms require target labels or remain inconclusive under the available probes (Section 2). We do not claim a universal law from four domains, nor that shift mechanism has been shown to outpredict shift magnitude; we claim a reproducible pattern of axis-specific failure, a magnitude-insufficiency finding, and a cheap protocol for testing both on a new domain. Establishing where mechanism-aware diagnosis does and does not add information beyond magnitude is the natural next step, and the benchmark is built to accept new domains.

What this evidence establishes, and what remains uncertain. Framed against the four hypotheses this paper set out to test: **H1** (different shift mechanisms produce distinguishable failure fingerprints) is *mixed* – only one of four possible mechanism-pair comparisons was both valid and informative enough to adjudicate. **H2a** (scalar shift magnitude alone is insufficient to characterize the observed failure variation) is *supported*, with both in-sample (447 cells, every axis) and out-of-sample (Section 4.3) evidence. **H2b** (shift mechanism carries information beyond magnitude about which axis fails) is *not established* – the direct test (mechanism-only vs. magnitude-only vs. both,

leave-one-out cross-validated) returns negative out-of-sample R^2 for every candidate model on the $n = 10$ mechanism-comparable points available, which we read as underpowered rather than as evidence against the hypothesis. **H3** (the staged audit is more actionable than a scalar score alone) is *not formally tested*; no comparison in this paper measures actionability directly. These four statuses, not a comparative mechanism-beats-magnitude claim, are this paper’s actual scientific position.

6 Related Work

Dataset-shift taxonomy. The covariate/prior/concept-shift decomposition TrustShift’s diagnosis protocol is built on is not new: it follows the canonical taxonomy of Moreno-Torres et al. [2012], the foundational treatment in Quiñero-Candela et al. [2009], and Shimodaira’s importance-weighting correction for covariate shift [Shimodaira, 2000]. TrustShift does not propose a new taxonomy; it operationalizes the existing one with label-free-where-possible probes and is explicit, via its *inconclusive* category, about where those probes cannot adjudicate.

Shift diagnosis and decomposition. Our probes build on black-box shift detection [Rabanser et al., 2019] and label-shift estimation [Lipton et al., 2018]. The domain-classifier probe (AUC_{dc}) is grounded in the $\mathcal{H}$ -divergence bound of Ben-David et al. [2010], which relates target error to source error plus a source/target divergence a classifier can estimate directly. The closest prior work is DISDE [Cai et al., 2025], which decomposes an aggregate performance *change* into harder-seen-inputs, $P(Y | X)$ -change, and new-input-region components with a formal estimator. DISDE answers *why aggregate performance moved*; TrustShift asks a different question – *which trustworthiness axis failed*, jointly across discrimination, calibration, operating-point performance, and subgroup reliability – and does not claim its categorical diagnosis explains failure better than a decomposition method such as DISDE would; that comparison is untested here.

Open-set, novel-class, and subpopulation shift. Genuine open-set recognition – new classes entering the label space at test time – is formalized by Scheirer et al. [2013]. Security’s deployment shift is deliberately *not* described in those terms: the binary task (benign vs. attack) is unchanged at test time, so no label-space expansion occurs. The closer prior-work anchor is subpopulation shift [Yang et al., 2023], which studies degradation on subpopulations absent or rare in training while the label space stays fixed – the same structure as security’s unseen attack families within the single positive class, and a framing we also draw on for the deployment-context result discussed in Section 5.

Shift robustness benchmarks such as WILDS [Koh et al., 2021] and TableShift [Gardner et al., 2023] quantify how much accuracy drops under shift, within a modality; Wiles et al. [2022] cross shift *type* with method across $\approx 85,000$ models and are the reason we do not claim prior work studies shift one domain or type at a time – that claim would be false and directly checkable against their own results. TrustShift differs from all three not in scale but in what it measures: calibration and subgroup reliability alongside discrimination, jointly, on a small number of real, non-synthetic, cross-modality deployment shifts, coupled to a staged diagnostic and remediation workflow rather than a robustness leaderboard.

Calibration under shift. Ovadia et al. [2019] establish that predictive-uncertainty calibration degrades under distribution shift and that post-hoc calibration methods can underperform in that

regime – the antecedent our own calibration finding (calibration and the operating point degrade far more than discrimination in three of four domains) builds on rather than rediscovers. Our post-hoc recalibration step uses temperature, Platt, and isotonic scaling [Guo et al., 2017, Zadrozny and Elkan, 2002]; we report Expected Calibration Error with equal-width binning and note, following Roelofs et al. [2022], that equal-mass binning has lower estimator bias – a sensitivity check this paper’s released configuration is built to run but has not yet executed. Calibration failure is not a cosmetic concern in a domain such as clinical risk prediction, where miscalibrated probabilities directly mislead decisions [Van Calster et al., 2019].

Subgroup and fairness reliability under shift has mostly been examined within single domains; Schrouff et al. [2022] show fairness properties fail to transfer in medical settings and that clinically plausible shifts violate the assumptions of single-mechanism mitigations. We corroborate the multi-mechanism point across four domains: the subgroup gap moves in different directions for different (model, shift) pairs within the same domain, not just across domains. A recent survey [Lin et al., 2024] catalogues methods; we contribute an empirical, cross-domain fingerprint rather than a new fairness-under-shift method.

Where TrustShift sits. TrustShift is not a new dataset-shift taxonomy – it uses the existing one. It is not a formal performance-decomposition estimator – it asks a different question than DISDE. It is not a claim that shift mechanism outpredicts shift magnitude – that direct comparison, run in Section 4.3, does not resolve in the affirmative. Its contribution is the joint discrimination/calibration/operating-point/subgroup-reliability audit, coupled to a staged, largely label-free diagnostic workflow that reports *inconclusive* as a first-class outcome, evaluated on four real, non-synthetic, cross-modality deployment shifts – a combination no single work cited above provides.

7 Limitations

Four domains and sixteen shift-points are evidence, not proof; the cross-domain regressions are exploratory synthesis over effectively four clusters, and we do not present them as a fitted predictor. Subgroup semantics differ across domains by necessity—demographic for clinical and lending, platform-derived proxy classes for text, and *operational* attack families for security—and we never interpret security’s subgroup reliability as demographic fairness. Clinical target-side subgroup analysis is limited to age: the released prediction arrays were generated from a previously harmonized BRFSS dataset whose preprocessing pipeline is no longer recoverable, so sex, race, and BMI analyses are not reproducible on the target side; age-based analyses remain fully reproducible and constitute the primary clinical subgroup evaluation. The clinical (source-only primary model, FedAvg) and lending subgroup gaps *narrow* under shift, which we attribute tentatively to changes in subgroup composition rather than to genuine fairness improvement, and flag as a hypothesis; this narrowing is model-dependent, not domain-dependent – the non-primary clinical boosted model’s gap *widens* under the identical shift ($\Delta G = +0.029$), so subgroup reliability here is a property of the (model, shift) pair. The security detector’s near-ceiling in-domain AUC (0.998) is measured under a capture-session-disjoint train/test split; a duplicate-row check found no evidence of excess cross-split duplication relative to within-split duplication (we do not claim this proves the split is leakage-free). A random row-level split of the same data (ignoring capture-file identity) gives a higher AUC of 1.000, reported alongside as a sensitivity comparison rather than as the primary number. We treat the native-split 0.998 as a genuine in-domain ceiling for whole-flow volumet-

ric/reflection features rather than a leakage artifact, though we have not yet formally characterized *why* the separability is so high. This study examines four deployment settings rather than an exhaustive sample of machine-learning domains; important modalities—especially vision foundation models and large language models, whose shift behaviour may differ—remain to be tested, and the benchmark is designed to accept them. Finally, all experiments run on a single workstation, bounding model scale.

Scope of the magnitude, context, and mechanism claims. The context-heterogeneity result (Section 4.3) uses a bare state identifier as the only context variable; no economic or regulatory state-level covariate was available in the current feature set, so we test whether population identity carries signal, not what about a population carries it. The magnitude-insufficiency claim (H2a) is evaluated against two proxies, AUC_{dc} and $|\Delta\pi|$; other severity proxies (e.g. population-stability index, raw score drift) are not tested. The direct mechanism-vs-magnitude comparison (H2b) has only $n = 10$ mechanism-comparable points after excluding cells without both a valid mechanism label and sufficient severity spread – underpowered, not merely inconclusive, and every candidate model in that comparison scores negative out-of-sample R^2 ; we do not treat this as evidence against H2b, only as evidence the current data cannot adjudicate it. H3 (staged diagnosis more actionable than a scalar score) has no formal test in this paper. The aggregation-induced reversal (Section 4.4) is demonstrated in lending only; whether the same weighting effect recurs in the other three domains is untested. Clinical has no covariate-shift probe at all – the only available proxy never reaches AUC_{dc} – so its *inconclusive* diagnosis reflects missing evidence, not evidence against covariate shift. NLP’s remediation-stage subgroup gap is undefined (NaN) for both targets in the released artifacts, a data-generation issue not yet root-caused; the remediation claim in Section 4.5 is scoped to clinical and lending’s subgroup-gap stability for this reason.

8 Conclusion

Deployment shift does not degrade machine-learning trustworthiness uniformly: discrimination, calibration, operating-point performance, and subgroup reliability can move independently, and even in different directions, within the same deployment. Scalar marginal shift scores – a domain-classifier AUC, a prevalence shift – were not sufficient, on their own, to explain which axis moved or by how much in the settings studied here. Deployment context can carry additional information beyond scalar magnitude when historical audit data for that population exist, though this signal does not extrapolate to a population never audited before. Mechanism-aware diagnosis is a promising organizing framework – it recovers a specific diagnosis for some shifts and honestly reports *inconclusive* for others – but its predictive superiority over scalar shift magnitude has not been established by the direct test this paper runs. In-domain metrics predicted none of the deployment outcomes we observed. We release the TrustShift protocol and benchmark so that new domains, and the open question of when mechanism adds information beyond magnitude, can be tested directly.

Statements and Declarations

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Competing interests. The author declares no competing interests.

Ethics approval. Not applicable. This study uses only publicly available, de-identified datasets and involves no new human- or animal-subjects data collection.

Author contributions. Rajveer Singh Pall is the sole author and conducted all aspects of the work: conception, methodology, experiments, analysis, and writing.

Data availability statement. All source datasets are public: NHANES and BRFSS (U.S. Centers for Disease Control and Prevention), HMDA (U.S. Consumer Financial Protection Bureau), GoEmotions, CIC-DDoS2019, and CICIDS2017. Consistent with their licenses, the released benchmark redistributes standardized model *predictions* and metadata—not raw third-party records. The harmonized benchmark of predictions is available on the Hugging Face Hub at <https://huggingface.co/datasets/Rajveer-code/trustshift>.

Code availability. The TrustShift protocol, the four domain adapters, the audit engine, and every results file that backs a number in this paper—together with a single script that regenerates all tables and figures from the committed prediction files—are available as an open-source repository at <https://github.com/Rajveer-code/trustshift>. Seeds, estimator settings, and metric definitions are fixed in the released configuration.

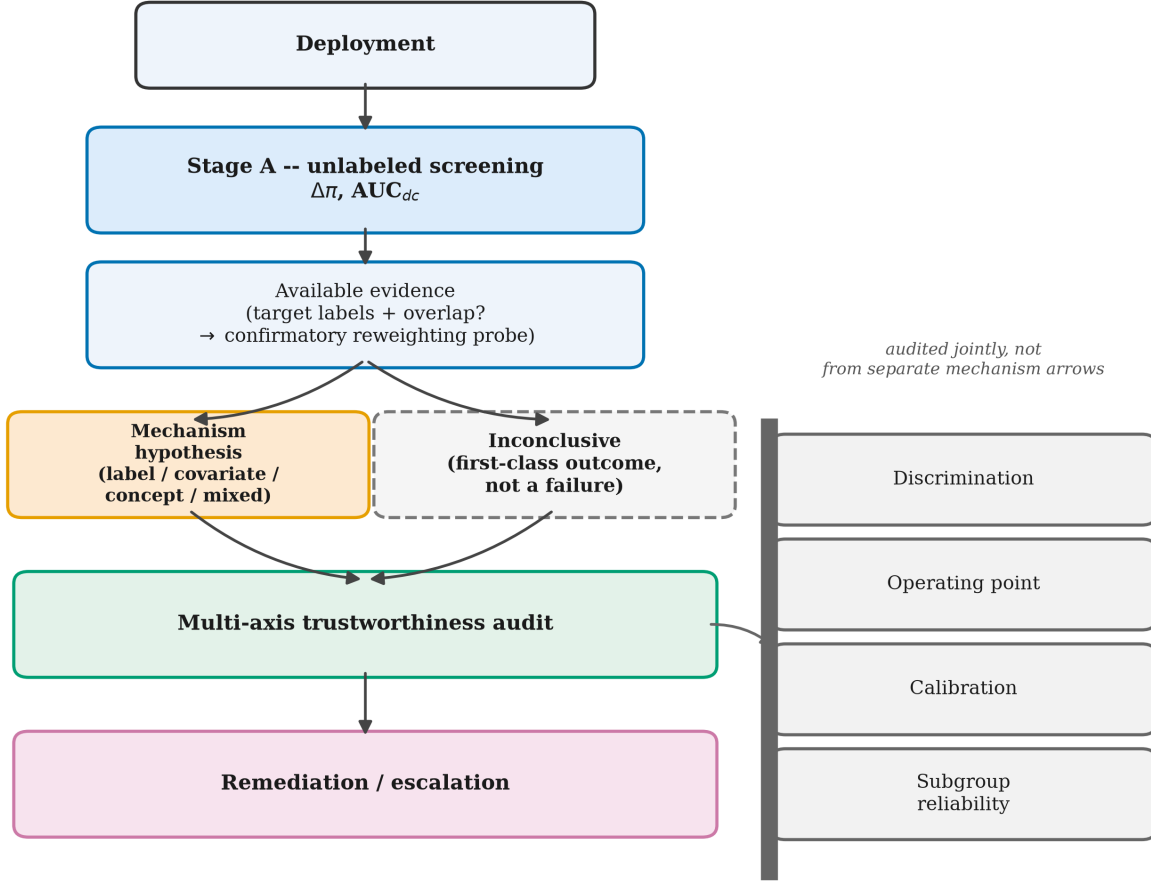
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The TrustShift audit flow



*Dashed border = evidence does not support a firm mechanism label.
No arrow in this figure implies a deterministic mechanism → failure-axis mapping.*

Figure 1: **The TrustShift audit flow.** A deployed model is first screened with cheap, label-free probes (Stage A); the available evidence – which depends on whether target labels and adequate source/target overlap exist – organizes either a mechanism hypothesis (label, covariate, concept, or mixed) or an explicit **inconclusive** outcome, shown with a dashed border because it is a first-class result, not a failure of the protocol. Both outcomes feed the same multi-axis trustworthiness audit, which jointly measures discrimination, operating-point performance, calibration, and subgroup reliability (right panel) before remediation or escalation. No arrow in this figure runs from an individual mechanism hypothesis to an individual trustworthiness axis: the four axes are audited jointly from whichever diagnosis is reached, and this paper’s own evidence (Section 4.3) does not establish that shift mechanism predicts which axis will fail.

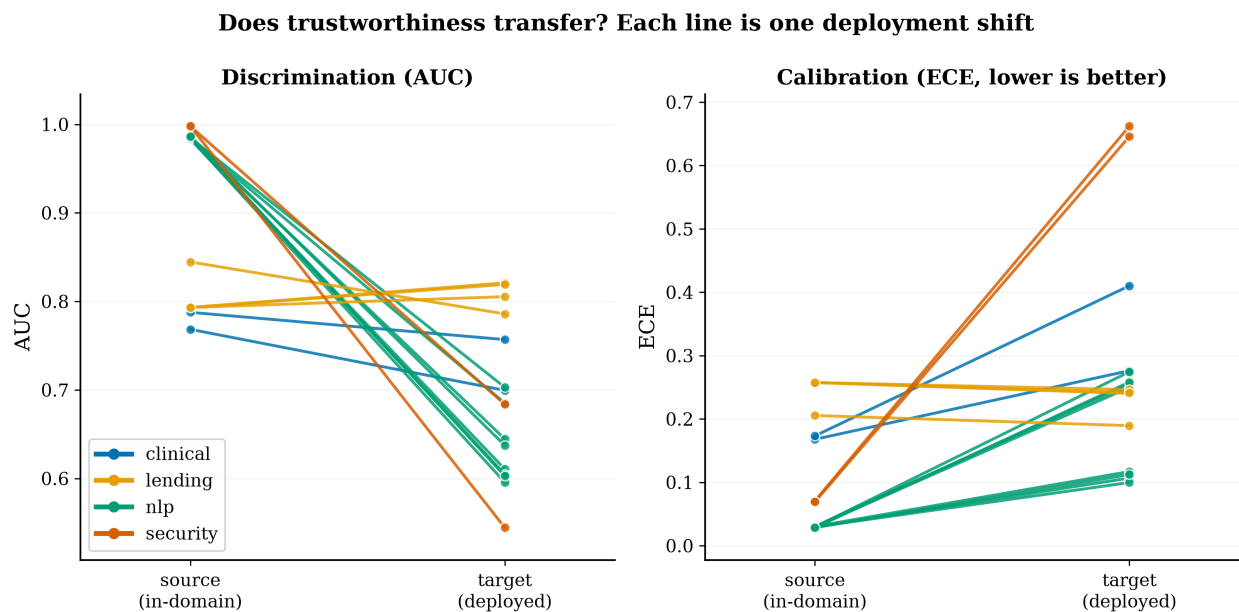


Figure 2: Discrimination (left) and calibration (right) as each shift transfers: every line runs from a shift-point’s source (in-domain) value to its target (deployed) value, colored by domain. Source values are high and uniform; target values fan out. Steep lines mark collapse (NLP, security AUC), flat lines robustness (lending). The clinical lines are near-flat for AUC yet rise steeply for ECE—calibration fails where discrimination does not.

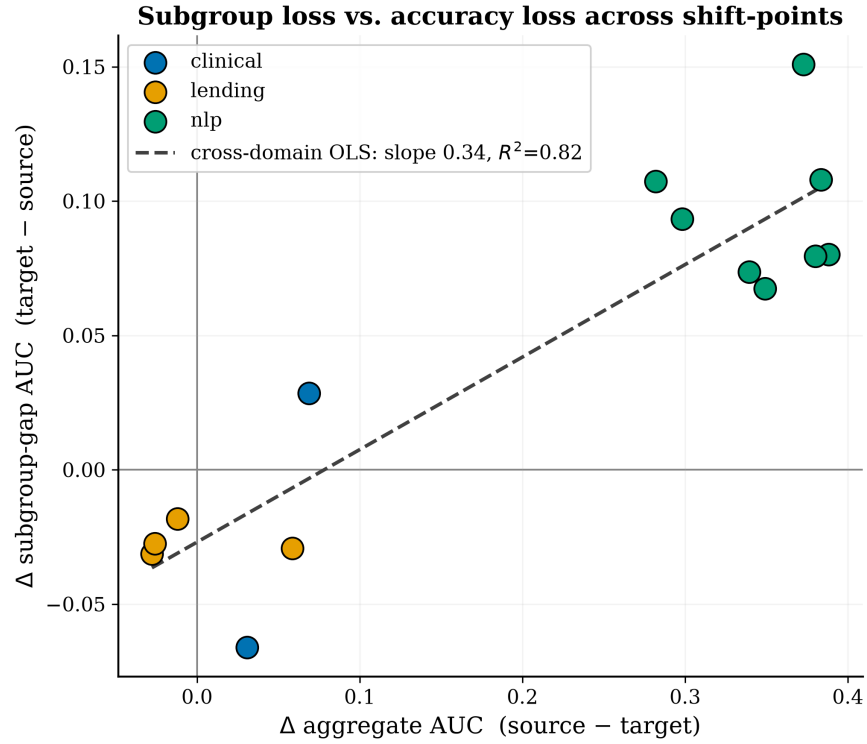


Figure 3: Subgroup-gap change vs. aggregate accuracy loss across shift-points, colored by domain. The cross-domain trend is real but domain-clustered; within-domain slopes differ (see box). Security is absent because its operational subgroups are all-positive, so a gap is undefined.